

Local RAG Development Environment Setup

Using LangChain, Ollama, HuggingFace and Visual Studio Code

1. Install Python

Download Python from <https://www.python.org/downloads/>

Run the installer and ensure 'Add Python to PATH' is checked.

Verify installation:

```
python --version
```

2. Install Visual Studio Code

Download VS Code from <https://code.visualstudio.com/>

Install with default settings.

3. Install Required VS Code Extensions

Open VS Code → Extensions (Ctrl+Shift+X) and install:

- Python
- Pylance
- Jupyter
- Python Environment Manager
- GitLens
- Markdown Preview (optional)
- Thunder Client (optional)

4. Install Ollama (Local LLM Runtime)

Download Ollama from <https://ollama.ai>

Verify installation:

```
ollama --version
```

Start Ollama:

ollama serve

Download a model:

ollama pull llama3.2:1b

5. Install Required Libraries

pip install langchain

pip install langchain-community

pip install langchain-text-splitters

pip install chromadb

pip install sentence-transformers

pip install pypdf

pip install unstructured

6. Configure VS Code Python Interpreter

Press Ctrl+Shift+P

Search: Python: Select Interpreter

Select ./venv

7. Create Sample Knowledge Base

Inside Knowledge folder create doc1.txt

Example content:

France is a country in Europe.

The capital of France is Paris.

8. Test Ollama from Python

Create test_llm.py

```
from langchain_community.chat_models import ChatOllama
```

```
llm = ChatOllama(model="llama3.2:1b")  
response = llm.invoke("What is the capital of France?")  
print(response)
```

9. Recommended Hardware

Minimum:

RAM: 8GB

CPU: 4 cores

Storage: 10GB

Recommended:

RAM: 16GB

CPU: 8 cores

GPU: Optional

13. Useful Ollama Commands

Run model:

```
ollama run llama3
```

List models:

```
ollama list
```

Remove model:

```
ollama rm llama3
```